

Week 11 - 11 December

Section outline

- - Date: **Thursday, 11 December**

Time: **14:00 - 18:00**

Location: **PR_B501D**

Lecturer: Zhou Zhou

Session Content

- Feature Visualisation
- Activity Detection (Vision Transformer, Huggingface)
- Assessment Submission Guideline
- Mini Project Planning

Week11 Slides Feature Visualization and Transformer Revolution

○ Resources:

Hugging Face (collection of models)

Universe (collection of computer vision datasets and APIs)

Models you might use:

Hand gesture detection: Yolov8x Tuned to Hand Gestures

Hand gesture models

Hand gesture continuously controlling: Mediapipe hand landmarks detection

Palmistry: Check the method they use to classify palm lines, and use it as the start for your project.

Datasets you might use:

Palm Lines Highlighted

Papers to check:

Palmistry using Machine Learning

Another Palmistry using Machine Learning

Additional resources:

Microgesture design for controlling things: [Hand tracking microgestures OpenXR extension](#)

Also could check [Wekinator](#), but don't use it for your projects cuz you are supposed to write code and understand the process of building a ML model.

○

Colab1 Feature VisualizationURL

Objective: Understand what a Convolutional Neural Network (CNN) actually "sees" at different stages of processing.

Concept: When a Deep Learning model looks at an image, it processes it through layers.

- Early Layers: Act like the retina. They detect simple lines, edges, and color gradients.
- Middle Layers: Combine edges to find shapes, circles, and patterns.
- Deep Layers: Combine shapes to identify complex concepts like "eyes," "wheels," or "fur."

In this exercise, we will use PyTorch Hooks. A "hook" allows us to intercept the data flowing through the network while it is processing an image, so we can save it and visualize it.

Instructions:

4. Run the setup code to load the model.
5. Run the visualization code.
6. Compare Layer 1 (Lines) vs. Layer 4 (Abstract Blobs).
7. Upload your own image to see how the network reacts to it.

○

Colab2 Object Detection with TransformersURL

In this notebook, we move away from CNNs and use a Transformer model called DETR (DEtection TRansformer).

Transformers were originally built for language (like ChatGPT), but they are now very powerful for computer vision. We will use the Hugging Face library, which makes using these complex models very easy.

What you will do:

- Install the transformers library.
- Load the DETR model pipeline.
- Detect objects in an image.
- Adjust the confidence threshold to filter out weak detections.

○

Colab3 Activity DetectionURL

Standard models can detect "Person" or "Cup". But can they detect "A person dancing" or "A broken cup"?

We will use CLIP, a model that understands the relationship between Images and Text. This allows for Zero-Shot Classification.

Zero-Shot means we do not need to train the model on specific categories. We can simply give it a list of text descriptions (labels), and it will tell us which one best matches the image.

What you will do:

- Load the Zero-Shot pipeline.
- Define your own custom labels (actions, moods, situations).
- Simulate a "Robot Brain" that makes a decision based on the detected activity.